WORKED SOLUTION

STEP III 2000, Question 11

Mechanics • Friction

STEP III 2000, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s3-q11

The key idea

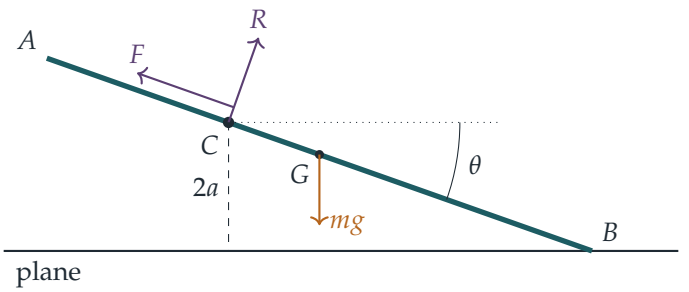
As long as there is no slipping, the rod simply pivots about the point C on the beam, so it is a rigid body rotating about a fixed axis. Energy gives the angular speed and the moment equation gives the angular acceleration. Then Newton's second law for the centre of mass, resolved along and across the rod, tells us the friction and normal reaction the beam must supply. Slipping is avoided while friction stays within μ times the normal reaction, and we need this to hold all the way until B reaches the plane.

Solution

The model and the geometry

We model the rod as a uniform rigid rod and the beam as a fixed thin rough line, so the beam exerts a single contact force at C: a normal reaction R perpendicular to the rod and a friction force F along it.

The centre of mass G is the midpoint, $4.5a$ from A. Since $AC = 3a$, G lies on the B side of C with $CG = \frac{3}{2}a$, and $CB = 6a$. So the weight has a moment about C, and the rod tips with B going down. Let θ be the angle the rod has turned through, so that the rod makes angle θ with the horizontal.



B reaches the plane when it has dropped $2a$, that is when $6a\sin\theta = 2a$, or

$$\sin\theta = \tfrac{1}{3}, \qquad \tan\theta = \frac{1}{2\sqrt{2}}.$$

The rotation about C

The moment of inertia about C is, by the parallel axes theorem,

$$I_C = \tfrac{1}{12}m(9a)^2 + m\left(\tfrac{3}{2}a\right)^2 = \tfrac{27}{4}ma^2 + \tfrac{9}{4}ma^2 = 9ma^2.$$

While there is no slipping, C is fixed and the contact force does no work. G has fallen $\frac{3}{2}a\sin\theta$, so conservation of energy (starting from rest) gives

$$\tfrac{1}{2}(9ma^2)\dot\theta^2 = mg\cdot\tfrac{3}{2}a\sin\theta \quad\Longrightarrow\quad \dot\theta^2 = \frac{g\sin\theta}{3a}.$$

Taking moments about C (only the weight has a moment):

$$9ma^2\,\ddot\theta = mg\cdot\tfrac{3}{2}a\cos\theta \quad\Longrightarrow\quad \ddot\theta = \frac{g\cos\theta}{6a}.$$

(Differentiating the energy equation with respect to t gives the same thing, a useful check.)

The contact forces

G moves on a circle of radius $\frac{3}{2}a$ about C, so its acceleration has a component $\frac{3}{2}a\dot\theta^2$ along the rod towards C and a component $\frac{3}{2}a\ddot\theta$ perpendicular to the rod, in the direction of increasing θ (downwards across the rod).

Ask yourself Which way does the friction act, and why is it not simply $mg\sin\theta$?

The component of the weight down the rod, $mg\sin\theta$, would pull the rod off the beam towards B; in addition G needs a centripetal force towards C. Both must be supplied by friction, which therefore acts up the rod, towards A. Resolving along the rod, taking the direction towards C as positive for the acceleration of G:

$$F - mg\sin\theta = m\cdot\tfrac{3}{2}a\dot\theta^2 = m\cdot\tfrac{3}{2}a\cdot\frac{g\sin\theta}{3a} = \tfrac{1}{2}mg\sin\theta,$$

so

$$F = \tfrac{3}{2}mg\sin\theta.$$

Resolving perpendicular to the rod, in the direction in which G is accelerating:

$$mg\cos\theta - R = m\cdot\tfrac{3}{2}a\ddot\theta = m\cdot\tfrac{3}{2}a\cdot\frac{g\cos\theta}{6a} = \tfrac{1}{4}mg\cos\theta,$$

so

$$R = \tfrac{3}{4}mg\cos\theta,$$

which is positive throughout, so the rod stays in contact with the beam.

The condition for no slipping

The rod does not slip at C as long as $F \le \mu R$, that is

$$\frac{F}{R} = \frac{\frac{3}{2}mg\sin\theta}{\frac{3}{4}mg\cos\theta} = 2\tan\theta \le \mu.$$

The ratio $2\tan\theta$ increases with θ, starting from 0 at release, so the condition is hardest to satisfy at the last moment, just as B reaches the plane. There $\tan\theta = 1/(2\sqrt{2})$, so we need

$$\mu \ge 2 \cdot \frac{1}{2\sqrt{2}} = \frac{1}{\sqrt{2}}.$$

Hence the minimum value of μ is

$$\boxed{\mu = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2} \approx 0.707.}$$

With exactly this value, friction becomes limiting only at the instant B strikes the plane; for any smaller μ slipping starts before then.

Checks

- At release $\dot\theta = 0$, so $F = 0$ and $R = \frac{3}{4}mg$: the beam carries less than the full weight because G starts accelerating downwards with $\frac{3}{2}a\ddot\theta = \frac{1}{4}g$. That is the familiar "a released plank feels lighter" effect.
- Units: F and R are both multiples of mg, and μ is dimensionless, as it must be.
- We stepped the equation $\ddot\theta = g\cos\theta/(6a)$ numerically from rest, computed the acceleration of G by finite differences, and recovered F and R from Newton's second law. They match $\frac{3}{2}mg\sin\theta$ and $\frac{3}{4}mg\cos\theta$, and F/R climbs steadily to $0.7071\ldots = 1/\sqrt{2}$ when $\sin\theta = \frac{1}{3}$.

Common slips

- Taking moments about the centre or forgetting the parallel axes term: $I_C = 9ma^2$, not $\frac{27}{4}ma^2$.
- Treating the problem as statics, $F = mg\sin\theta$, $R = mg\cos\theta$, which gives $\mu \ge \tan\theta$. Both the centripetal and the tangential acceleration of G change the forces.
- Checking the friction condition only at release, where $F = 0$, rather than at the point where it is most demanding.

Beyond STEP

This question is a first taste of rigid-body dynamics with constraint forces: we found the motion from energy and moments, and only then used Newton's law for the centre

of mass to work backwards to the forces the constraint must provide. The same pattern (solve the motion, then check whether the constraint can actually supply the force needed) decides when a ladder starts to slip, when a toppling pencil leaves the table, or when a bead leaves a wire. Kleppner and Kolenkow, *An Introduction to Mechanics*, develops fixed-axis rotation and the link between torque and the motion of the centre of mass carefully.

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